

ABCDs of Nutritional Evaluation

Part D: Diet and Lifestyle Review



ELIZABETH BOHAM, MD, RD

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Elizabeth Boham, MD, RD, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. Recall and describe two point-of-care clinical tools designed to facilitate the dietary review of patients.
2. Explain how these tools incorporate data from anthropometrics, biomarkers, clinical exam findings, and other contextual features of the patient's lifestyle.

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Utilize two point-of-care clinical tools to evaluate a patient's dietary intake in the context of their lifestyle, anthropometrics, biomarkers, and clinical findings, to assess and estimate health risks, and to identify areas of opportunity for dietary interventions.

Clinical Focus

- Point-of-care clinical tools for the functional nutrition evaluation: **Diet, Nutrition, & Lifestyle Journal**



ABCDs of the Functional Nutrition Evaluation



Anthropometrics
& Vital Signs



Biomarkers &
Functional Labs

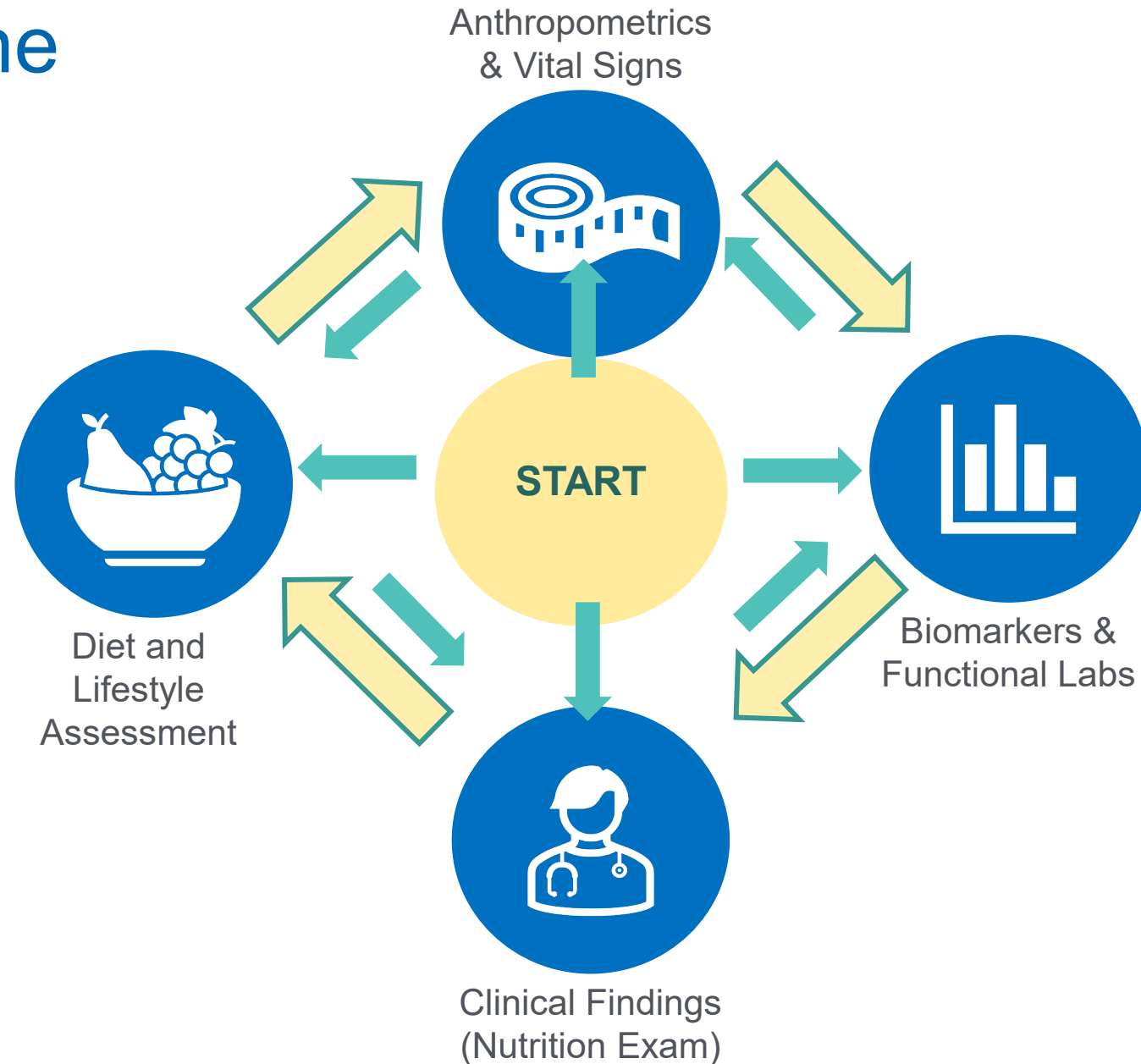


Clinical Findings
(Nutrition Exam)



Diet and Lifestyle
Assessment

ABCDs of the Functional Nutrition Evaluation



Long description: A B C Ds of the Functional Nutrition Evaluation

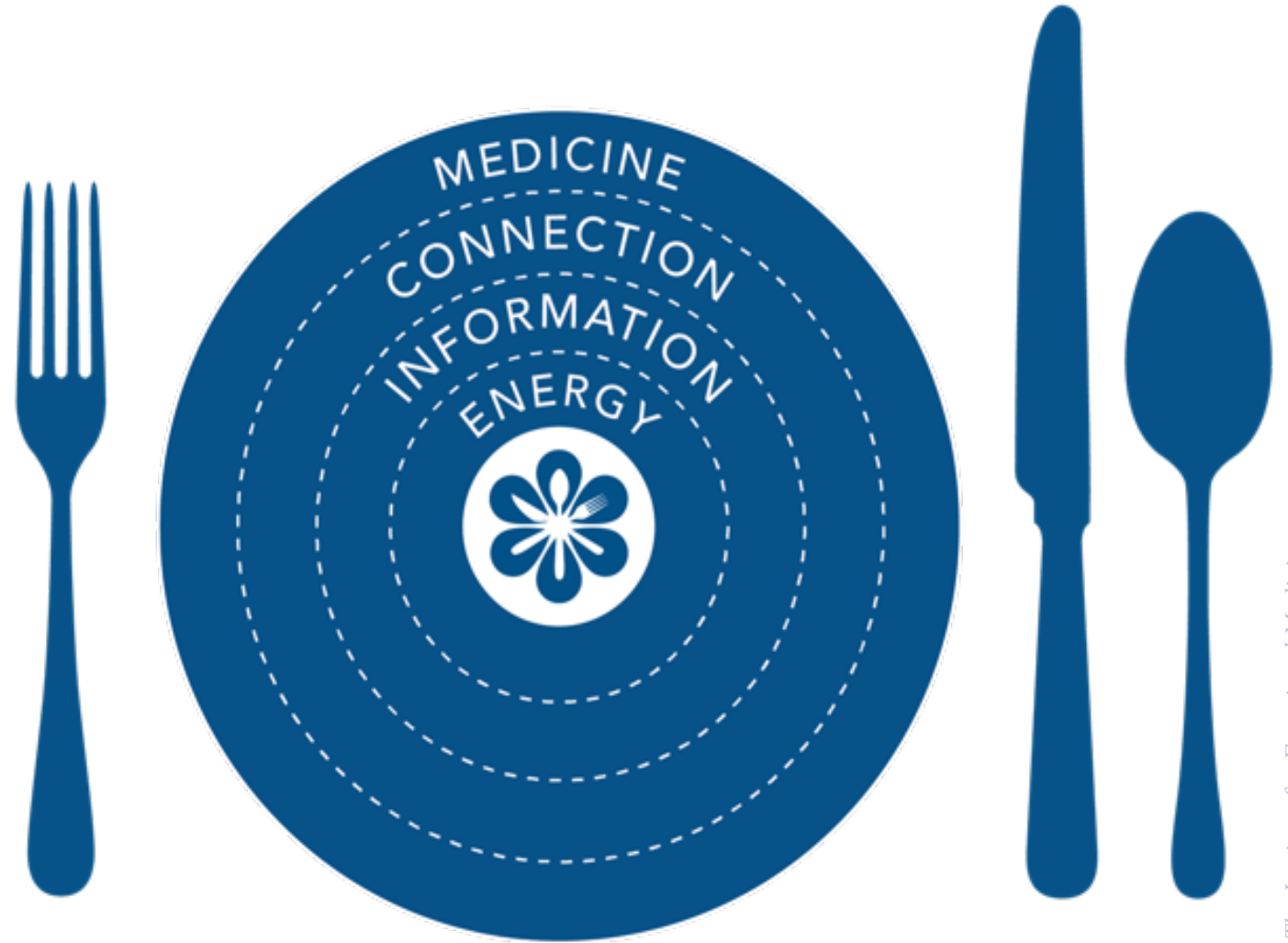
A graphic depicting a sequence:

- **Anthropometrics and Vital Signs:** Involves body measurements and health indicators.
- **Biomarkers and Functional Labs:** Involves testing biological markers for health assessment.
- **Clinical Findings (Nutrition Exam):** Involves observations from a nutrition-focused physical exam.
- **Diet and Lifestyle Assessment:** Involves evaluating dietary habits and lifestyle factors.

The graphic suggests that these elements are interconnected steps in a comprehensive nutritional evaluation process.

Food Is...

- Medicine
- Connection
- Information
- Energy



The Clinical Tools of the Functional Nutrition Evaluation **Diet, Nutrition, & Lifestyle Journal**

Diet, Nutrition & Lifestyle Journal—1 Day

Patient Name
Date
Food Plan Type

Day 1

Day Event	Food & Drink Intake (include type, amount, brand)	Macronutrients (PFC) and Phytonutrients
Rising Time		P F C ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Breakfast Time		P F C ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
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Dinner Time		P F C ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
PM Snack Time		P F C ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Bed Time		P F C ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR

P: Proteins; F: Fats; C: Carbohydrates; R: Red; O: Orange; Y: Yellow; G: Green; B/P/BL: Blue/Purple/Black; W/T/BR: White/Tan/Brown

Sleep & Relaxation	Exercise & Movement	Stress	Relationships
Sleep Quantity: (hrs) Quality: ☐ Poor ☐ Fair ☐ Good Relaxation ☐ Yes ☐ No Type/Amount: 	Type, Duration, & Intensity ☐ Aerobic: ☐ Strength: ☐ Flexibility: 	Stress reduction practices: Stressors: 	Supporting: Non-supporting:

Mental	Emotional	Spiritual

The Diet, Nutrition, & Lifestyle Journal

- Identify dietary patterns and habits
- Identify food and beverage choices
- Correlate dietary patterns with circadian lifestyle behaviors
- Correlate dietary patterns with mental, emotional, and spiritual factors

Diet, Nutrition & Lifestyle Journal—1 Day

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Mental	Emotional	Spiritual

Food and Beverages

Modifiable Lifestyle Factors

Mental, Emotional, Spiritual

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Mental	Emotional	Spiritual

Food and Beverages

Modifiable Lifestyle Factors

Mental, Emotional, Spiritual

Day 1

Day Event	Food & Drink Intake (include type, amount, brand)	Macronutrients (PFC) and Phytonutrients
Rising Time 7 am		P F C ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Breakfast Time 8 am	Oatmeal, toast & butter, coffee	P ★ F ★ ★ C ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Mid-AM Snack Time 11 am	Muffin, orange juice	P F ★ ★ C ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Lunch Time 1 pm	Macaroni and cheese, apple, milk	★ P F ★ ★ C ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Mid-PM Snack Time 4 pm	Potato chips and soda	P F ★ ★ C ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Dinner Time 7 pm	Pizza, salad, beer, cake	★ P ★ ★ F ★ ★ C ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
PM Snack Time 10 pm	Popcorn and soda	P F ★ ★ C ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Bed Time 11 pm		P F C ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR

P: Proteins; F: Fats; C: Carbohydrates; R: Red; O: Orange; Y: Yellow; G: Green; B/P/BL: Blue/Purple/Black; W/T/BR: White/Tan/Brown

Long Description: Diet, Nutrition, and Lifestyle Journal Example

Close up of the day 1 journal. The following information is covered in each line:

Macronutrients and phytonutrients for each meal are indicated by stars and various colored marks on the far right.

7am Rising time

8am breakfast time: Oatmeal, toast and butter, coffee

11am Muffin, orange juice

1pm Macaroni and cheese, apple, milk

4pm potato chips and soda

7pm Pizza, salad, beer, cake

10pm popcorn and soda

11pm Bedtime

Diet, Nutrition & Lifestyle Journal—1 Day

Patient Name
Date
Food Plan Type

Day 1

Day Event	Food & Drink Intake (include type, amount, brand)	Macronutrients (PFC) and Phytonutrients
Rising Time		F C ☐ R ☐ O ☐ Y ☐ G ☐ B/F/BL ☐ W/T/BR
Breakfast Time		F C ☐ R ☐ O ☐ Y ☐ G ☐ B/F/BL ☐ W/T/BR
Mid-AM Snack Time		F C ☐ R ☐ O ☐ Y ☐ G ☐ B/F/BL ☐ W/T/BR
Lunch Time		F C ☐ R ☐ O ☐ Y ☐ G ☐ B/F/BL ☐ W/T/BR
Mid-PM Snack Time		F C ☐ R ☐ O ☐ Y ☐ G ☐ B/F/BL ☐ W/T/BR
Dinner Time		F C ☐ R ☐ O ☐ Y ☐ G ☐ B/F/BL ☐ W/T/BR
PM Snack Time		F C ☐ R ☐ O ☐ Y ☐ G ☐ B/F/BL ☐ W/T/BR
Bed Time		F C ☐ R ☐ O ☐ Y ☐ G ☐ B/F/BL ☐ W/T/BR

P: Proteins; F: Fats; C: Carbohydrates; R: Red; O: Orange; Y: Yellow; G: Green; B/F/BL: blue/Purple/black; W/T/BR: White/tan/brown

Sleep & Relaxation	Exercise & Movement	Stress	Relationships
Sleep Quantity: (hrs) Quality: ☐ Poor ☐ Fair ☐ Good Relaxation ☐ Yes ☐ No Type/Amount: 	Type, Duration, & Intensity ☐ Aerobic: ☐ Strength: ☐ Flexibility: 	Stress reduction practices: Stressors: 	Supporting: Non-supporting:

Mental	Emotional	Spiritual

Food and Beverages

Modifiable Lifestyle Factors

Mental, Emotional, Spiritual

Sleep & Relaxation	Exercise & Movement	Stress	Relationships
Sleep Quantity 8 (hrs) Quality: ☐ Poor ☒ Fair ☐ Good Relaxation ☒ Yes ☐ No Type/Amount: TV before bed	Type, Duration, & Intensity Aerobic: Walked the dog ☐ Strength: None ☐ Flexibility: None	Stress reduction practices: None Stressors: Deadlines at work	Supporting: Dog Non-supporting: Supervisor angry

Mental	Emotional	Spiritual
"Sluggish"	"Tense"	"Prayed for strength"

Long Description: D N L M L F Example

Close up of the modifiable lifestyle factors section of the diet nutrition and lifestyle journal. The following information is provided:

- Sleep quantity: 8 hours, fair quality.
- Relaxation activities: yes; TV before bed
- Exercise and movement: Aerobic walked the dog, no strength or flexibility activities.
- Stress reduction: none.
- Stressors: deadlines at work.
- Supporting relationships: dog.
- Non-supporting relationships: angry supervisor.
- Mental: Sluggish; Emotional: Tense; Spiritual: Prayed for strength

Clinical Focus

- Point-of-care clinical tools for the functional nutrition evaluation: **Organizing the Functional Nutrition Evaluation**



The Clinical Tools of the Functional Nutrition Evaluation **Organizing the Functional Nutrition Evaluation**

Organizing the Functional Nutrition Evaluation

Case Example

- 43 yo, Female
- Weight: 180 lbs.
- Height: 5'6"
- BMI: 29 (*overweight*)
- Waist circumference – *elevated* (>35 inch)
- Waist-to-hip – *elevated* (>0.85)



Organizing the Functional Nutrition Evaluation Point-of-Care Tool

THE INSTITUTE FOR FUNCTIONAL MEDICINE[®] Organizing the Functional Nutrition Evaluation						
	Proteins	Fats and Oils	Carbohydrates	Minerals	Vitamins	Phytonutrients
Medical History						
Anthropometrics						
Biomarkers						
Clinical Exam						
Dietary Review						

Organizing the Functional Nutrition Evaluation Case Example

 Organizing the Functional Nutrition Evaluation						
	Proteins	Fats and Oils	Carbohydrates	Minerals	Vitamins	Phytonutrients
Medical History	Vegetarian x 20 yrs	Margarine & trans-fats	Baked goods	Vegetarian x 20 yrs	Vegetarian x 20 yrs	Rarely veggies
Anthropometrics						
Biomarkers						
Clinical Exam						
Dietary Review						

Organizing the Functional Nutrition Evaluation

Case Example

 Organizing the Functional Nutrition Evaluation						
	Proteins	Fats and Oils	Carbohydrates	Minerals	Vitamins	Phytonutrients
Medical History	Vegetarian x 20 yrs	Margarine & trans-fats	Baked goods	Vegetarian x 20 yrs	Vegetarian x 20 yrs	Rarely veggies
Anthropometrics	5' 6" 167 cm	180 lb 82 kg	BMI 29	37" WC 94 cm	41" HC 104 cm	WHR 0.90
Biomarkers						
Clinical Exam						
Dietary Review						

Organizing the Functional Nutrition Evaluation

Case Example

 Organizing the Functional Nutrition Evaluation						
	Proteins	Fats and Oils	Carbohydrates	Minerals	Vitamins	Phytonutrients
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Anthropometrics	5' 6" 167 cm	180 lb 82 kg	BMI 29	37" WC 94 cm	41" HC 104 cm	WHR 0.90
Biomarkers	Low Protein & Albumin	High LDL Low Omega 3:6 Ratio*	High Trigs TG/HDL= 5 FBS = 110	Low Hgb/Hct/RBC Alk Phos	High MCV Low WBC High ALT AST	Requires advanced testing
Clinical Exam						
Dietary Review						

Organizing the Functional Nutrition Evaluation

Case Example

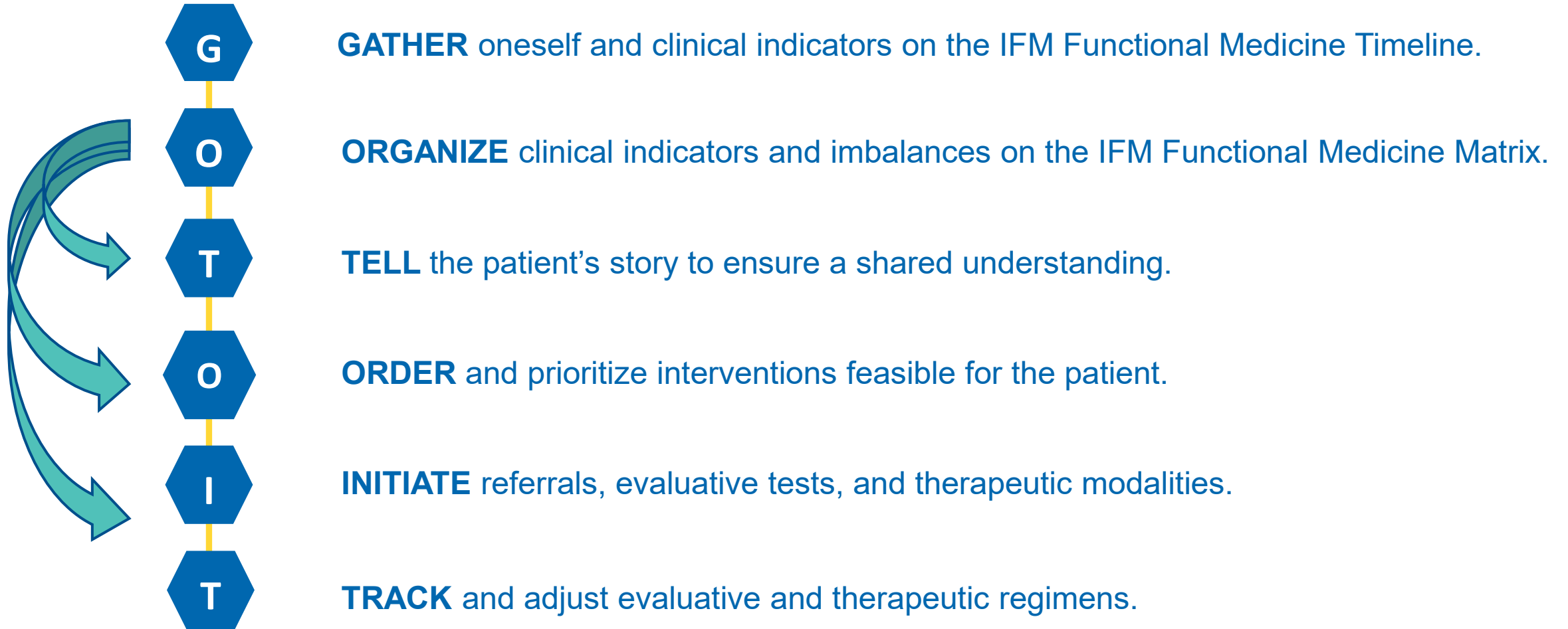
 Organizing the Functional Nutrition Evaluation						
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Biomarkers	Low Protein & Albumin	High LDL Low Omega 3:6 Ratio*	High Trigs TG/HDL= 5 FBS = 110	Low Hgb/Hct/RBC Alk Phos	High MCV Low WBC Plts ALT AST	Requires advanced testing
Clinical Exam	Thinning hair Brittle nails Glossitis	Xerosis Hyper- keratosis	Caries Acanthosis nigricans	Brittle nails Leukonychia Koilonychia	Aphthous stomatitis Enamel hypo	Often aligned w/vitamin & mineral signs
Dietary Review						

Organizing the Functional Nutrition Evaluation

Case Example

 Organizing the Functional Nutrition Evaluation						
	Proteins	Fats and Oils	Carbohydrates	Minerals	Vitamins	Phytonutrients
Medical History	Vegetarian x 20 yrs	Margarine & trans-fats	Baked goods	Vegetarian x 20 yrs	Vegetarian x 20 yrs	Rarely veggies
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Dietary Review	Processed soy No eggs	Canola Sesame Corn	Baked goods Pasta, Rice Juice, Soda	Diet limited No supplements	Diet limited No supplements	Limited to salads Corn, Potato

GOTOIT Functional Medicine Framework



Tools to help INITIATE a plan

THE INSTITUTE FOR
FUNCTIONAL MEDICINE®

Functional Medicine Prescription & Lifestyle Plan

Patient Name _____ Date of Birth _____

Functional Nutrition Plan

Foundational Interventions

☐ Phytonutrient Spectrum

☐ Core Food Plan

☐ Vegan

☐ Vegetarian

First-Step Interventions

☐ Cardiometabolic Food Plan

☐ Elimination Diet

☐ Food Reintroduction

Advanced Therapeutic Interventions

☐ Anti-Candida Food Plan

☐ Detox Food Plan

☐ Low-FODMAP Food Plan

☐ Mito Food Plan

Personal Dietary Recommendations

Macronutrient Distribution (protein/fat/carbs): ☐ 20/30/50 ☐ 25/30/45 ☐ 30/30/40 ☐ 20/60/20

Target Calories: ☐ 1000-1200 ☐ 1200-1400 ☐ 1400-1800 ☐ 1800-2200 ☐ 2200-2500

Intermittent Fasting: ☐ Yes ☐ No Target calories per day: _____ Frequency: ____ times per week

Other Recommendations: _____

Lifestyle Plan

Sleep

Recommendations: _____

Exercise

Risk Assessment: ☐ Low Risk ☐ Medium Risk ☐ High Risk

Clearance: ☐ Yes ☐ No _____

Exercise Prescription:	Cardio/Aerobic	Strength/Resistance	Flexibility/Stretching	Balance
F-Frequency (times per week)				
I-Intensity (e.g., low, moderate, vigorous)				
T-Time/Duration (minutes each day)				
T-Type (walking, jogging, etc.)				

Restoration

☐ Self-Awareness/Mindfulness ☐ Guided Imagery/Visualization ☐ Relaxation Response

☐ Breathing Techniques ☐ Meditation ☐ Other: _____

Disclaimer: This form is intended to be used within the scope of your practice.

Date: _____

Clinical Focus

- Conclusion, summary, and key takeaways.



ABCDs of the Functional Nutrition Evaluation



Anthropometrics
& Vital Signs



Biomarkers &
Functional Labs



Clinical Findings
(Nutrition Exam)



Diet and Lifestyle
Assessment

Nutritional Evaluation Diet & Lifestyle: Key Clinical Takeaways



In the “D” (Dietary Evaluation) of the ABCD evaluation, two point-of-care clinical tools have excellent clinical utility for organizing data related to anthropometrics, biomarkers, clinical findings, and dietary intake.

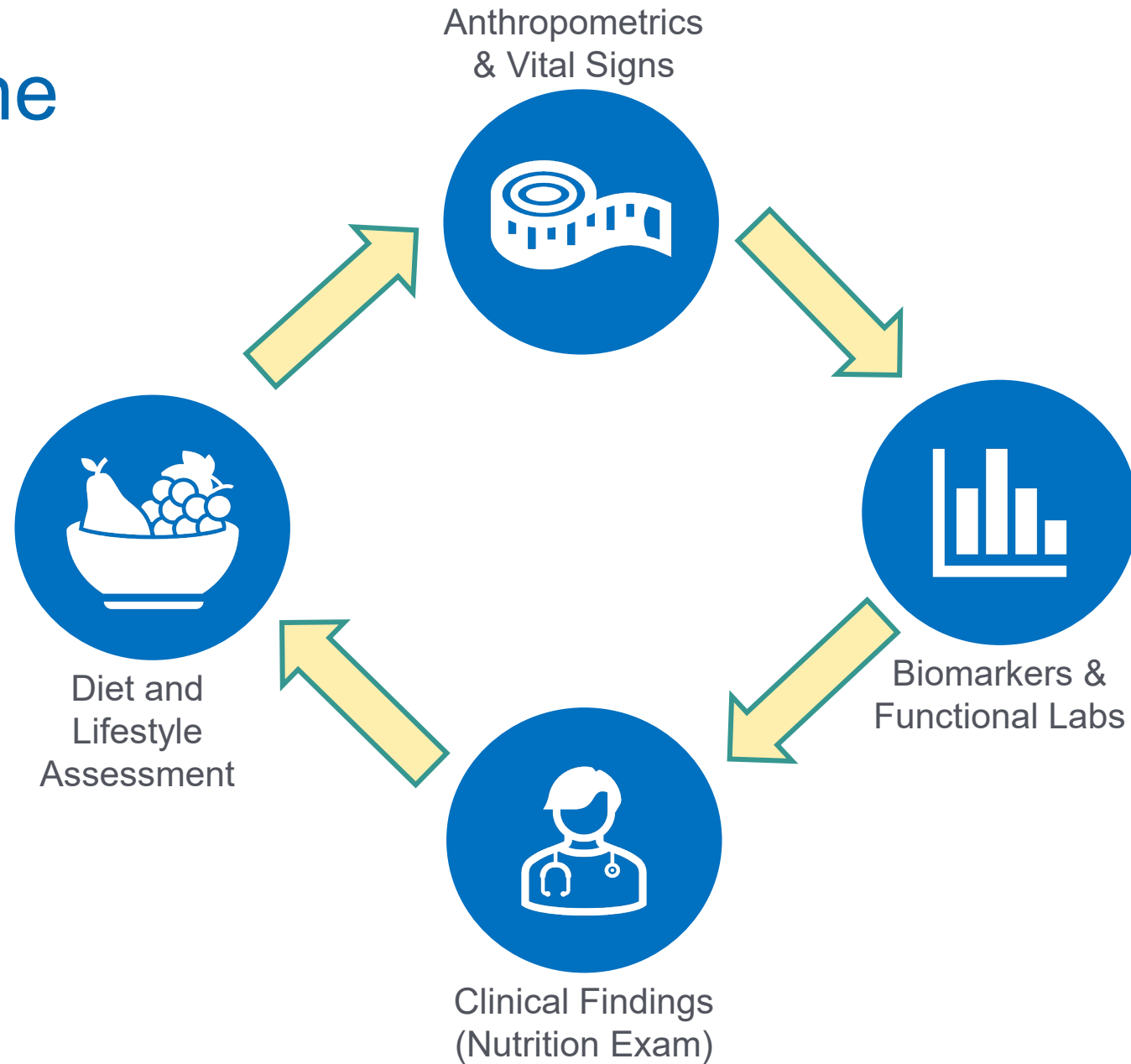


The evidence regarding the estimation of a patient’s health risks based on anthropometric measurements is emerging and evolving.

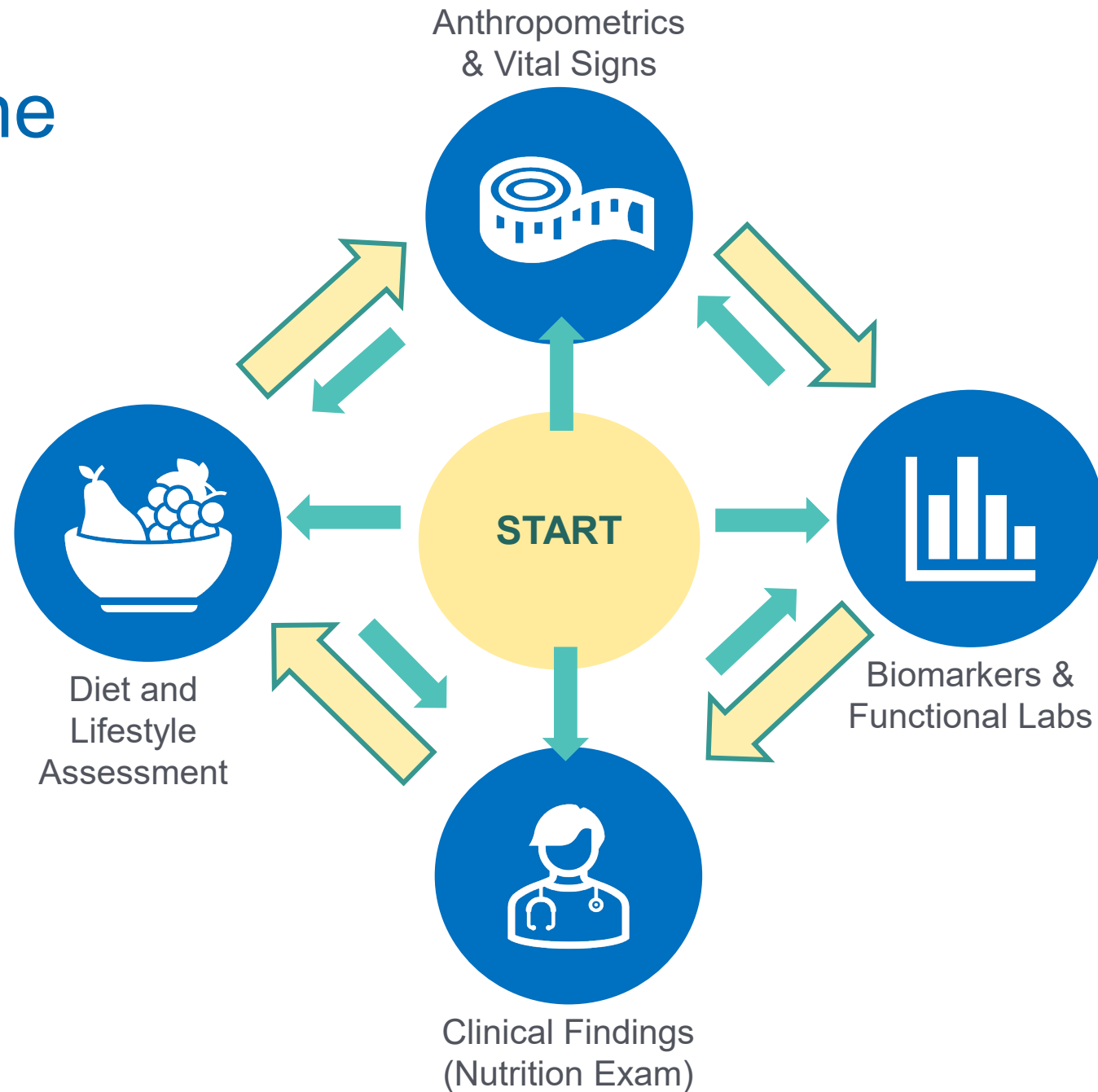


A complete functional nutrition evaluation applying the ABCDs can assist clinicians in identifying areas of greatest opportunity for dietary interventions most likely to improve a patient’s risk profile.

ABCDs of the Functional Nutrition Evaluation



ABCDs of the Functional Nutrition Evaluation



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Sleep Quantity (hrs)	Type, Duration, & Intensity ☐ Aerobic:	Stress reduction practices:	Supporting:
Quality: ☐ Poor ☐ Fair ☐ Good			
Relaxation ☐ Yes ☐ No	☐ Strength:	Stressors:	Non-supporting:
Type/Amount: 			
	☐ Flexibility:		

Mental	Emotional	Spiritual

The Diet, Nutrition, & Lifestyle Journal

- Identify dietary patterns and habits
- Identify food and beverage choices
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- Correlate dietary patterns with mental, emotional, and spiritual factors

Accessible version available in your toolkit.

Organizing the Functional Nutrition Evaluation Point-of-Care Tool

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Anthropometrics						
Biomarkers						
Clinical Exam						
Dietary Review						

ABCDs of Nutritional Evaluation

Part D: Diet and Lifestyle Review

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE